

Second CERP



Design Methodology of Optimized and Secured Hardware IPs

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**PSO based exploration of multi-phase encryption
based secured image processing filter hardware IP core
datapath during high level synthesis**

Image processing filters

- Image processing filters are mainly used to suppress either the high frequencies in the image, *i.e.*, smoothing the image, or the low frequencies, *i.e.*, enhancing or detecting edges in the image.
- The main objective of image processing is to extract some useful information from an image.
- From detection and recognition of license plates of vehicles on tolls (character recognition), advanced medical imagery (image analysis), biometric fingerprinting, robotics vision, and military operations to car driving automation, image processing plays a crucial role everywhere.
- Due to globalization of design supply chain, the design process of these image processing filters as a dedicated intellectual property (IP) core involves various hardware threats [1]- [4].

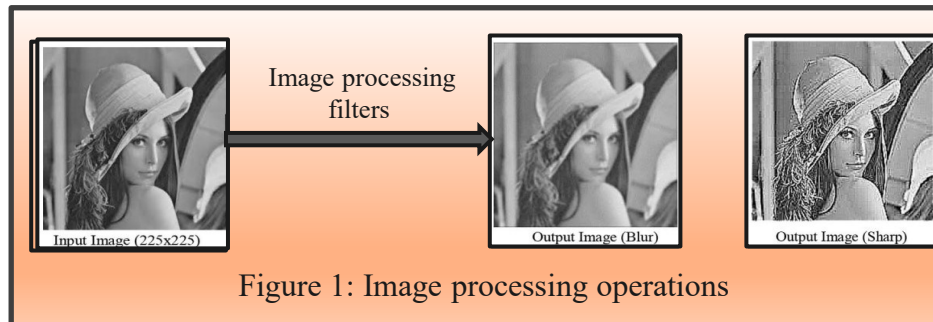
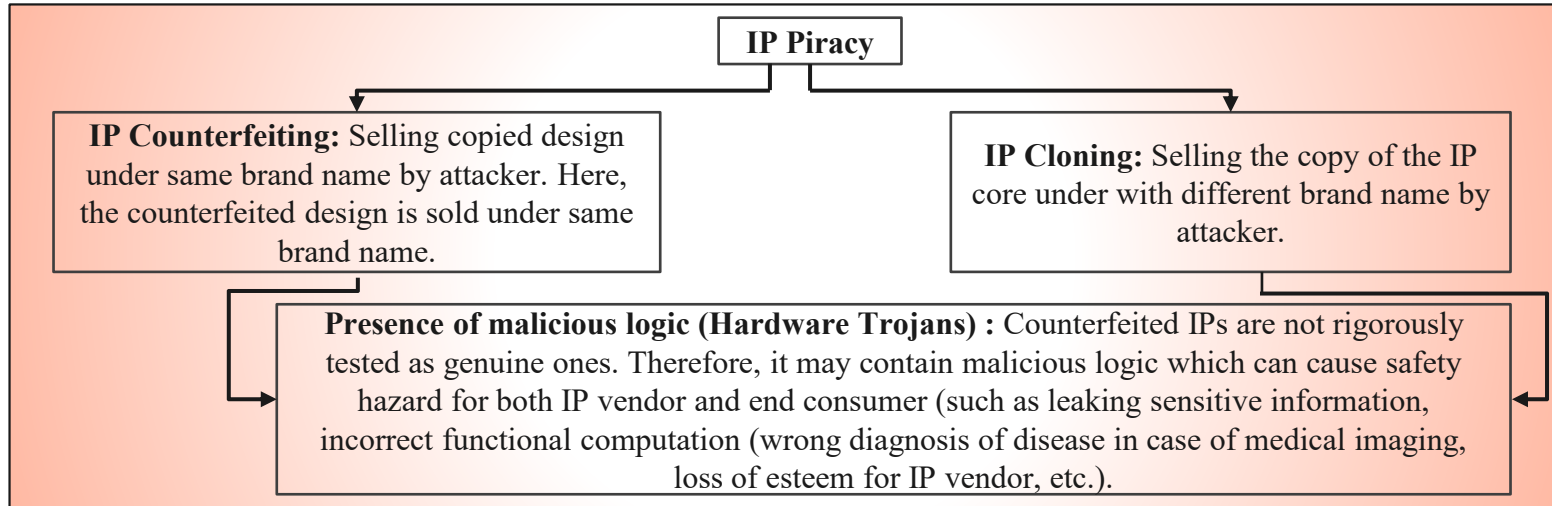


Figure 1: Image processing operations

Security Issues associated with image processing filter IP Cores [1], [2]:



Fraudulent claim of IP ownership: An adversary tries to fraudulently claim the ownership of the IP.

Therefore, it is essential to secure these image processing filter IP cores from these hardware threats.

Related Work :

Sr. No.	Existing Work	Technique Used	Remarks
1.	D. Tsiktiris, D. Ziouzos, and M.Dasygenis [5] (2018)	Authors discusses about the implementation of FPGA based image processing accelerators.	Does not focus on the security aspects of image processing filter hardware IPs.
2.	C. Shu, W. Pang, H. Liu, and S. Lu [6] (2019)	The paper focuses on designing of hardware accelerators for performing convolutional neural network (CNN).	Does not provide a framework for designing image processing filter IP cores using high level synthesis. Further, it also does not includes the security aspects of image processing filter hardware IPs.
3.	F. Koushanfar, I. Hong, and M. Potkonjak [7] (2005)	Hardware watermarking using two-variable (0, 1) signature encoding process.	Weak watermarking mechanism due to involvement of only two variable signature encoding process. Not robust and future proof. The watermark (original signature) inserted by watermarking technique becomes vulnerable if relevant information (like signature size, digit encoding, and digit combination) gets leaked.

Related Work :

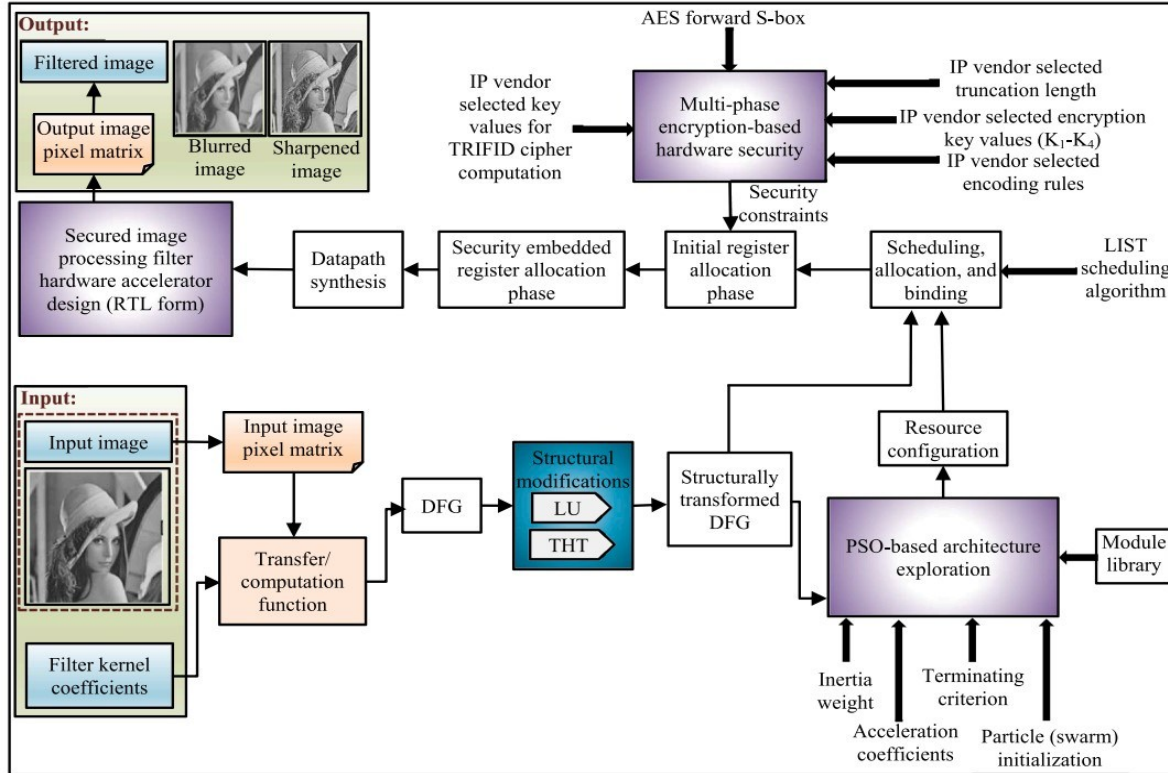
Sr. No.	Existing Work	Technique Used	Remarks
4.	A. Sengupta and M. Rathor [8] (2019)	Hardware steganography based security approach to address the IP counterfeiting threat using IP vendor specified threshold entropy value.	Signature free, becomes weak if secret value of chosen entropy threshold are leaked.
5.	A. Sengupta and S. Bhadauria [9] (2016)	Exploring low cost optimal watermark for reusable IP cores during high level synthesis using quadruple variable watermarking.	Watermark Signature becomes vulnerable if digit encoding and signature size gets leaked.

Proposed Work [3]:



- The proposed approach based on multi-phase encryption technique used for securing image processing filters IP cores. This paper discusses signature-based hardware security methodology on image processing IP cores for the first time.
- The proposed approach uses the register allocation table of the image processing application to generate secret data, which is used for multi-phase encryption to determine hardware security constraints .
- The generated hardware security constraints then embedded in the image processing filter IP cores design to authenticate genuine IP Maker.
- The huge variation in the key selection in different phases of encryption and the use of secret steganographic data for signature generation increases the robustness of the proposed hardware security methodology for image filters.

Detailed flow diagram of the proposed approach [3]:



[3] Aditya Anshul, Anirban Sengupta, PSO based exploration of multi-phase encryption based secured image processing filter hardware IP core datapath during high level synthesis, *Elsevier Expert Systems with Applications*, Volume 223, 2023, 119927.

Figure 2: Figure illustrating the overview of the proposed low-cost multi-phase encryption based hardware security for securing image processing filter IP cores

Particle swarm optimization-driven design space exploration (PSO-DSE):

- The integration of the PSO-DSE block with the encryption based security methodology serves the objective of determining an optimized architectural solution.
- PSO prunes design search space based on IP vendor specified orthogonal parameters such as area, delay, energy, power, etc. corresponding to secured image filter design to generate an optimized low-cost design.

Advantage of PSO-DSE [10] over others such as genetic algorithm [11] and bacterial foraging [12] based DSE:

- PSO-DSE considers the magnitude of the previously computed velocity with the help of a parameter called inertia weight, while [11] and [12] do not consider the momentum of prior iterations, which increases the probability of getting stuck in the local minima during architecture exploration.
- PSO-DSE creates a balance between exploitation and exploration time with the help of linearly decreasing the value of inertia from 0.9 to 0.1. The algorithm takes more significant steps at the beginning and smaller steps on reaching higher fitness solutions, which is missing in [11] and [12]. This also enhances the chance of reaching global optimal solution.
- The inclusion of various other factors (hyperparameters), such as social and cognitive factors in PSO-DSE, helps achieve higher fitness solution within a very low exploration time.

Details of multi-phase encryption based hardware security methodology:

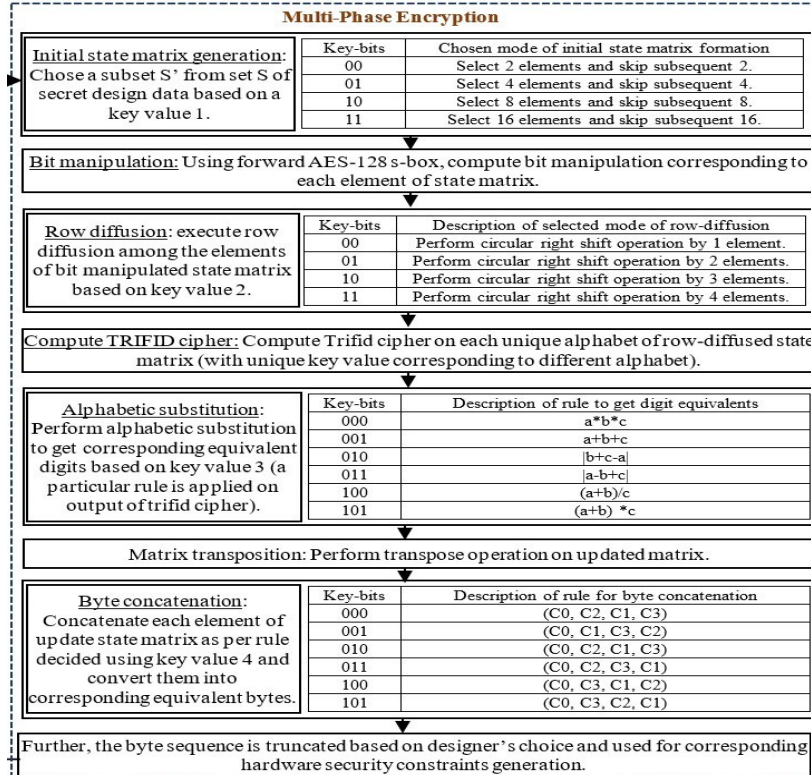


Figure 3: Flow-chart of proposed methodology for securing image processing filters using multi-phase encryption

Extraction of secret design data from SDFG of image processing application:

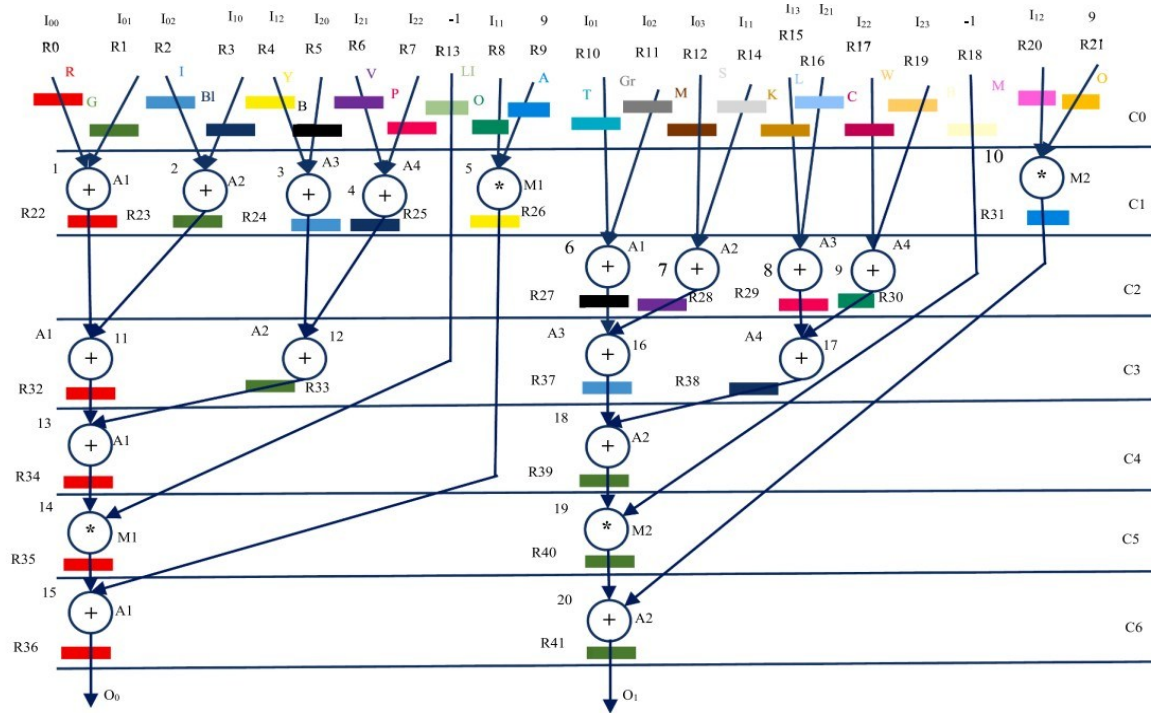


Figure 4: PSO-based architecture exploration of 4(+), and 2(*) resources used for scheduling 3*3 sharpening filter

Determination of secret design data based on scheduled data flow of image processing application:

Table 1: Register allocation table corresponding to 3*3 sharpening filter:

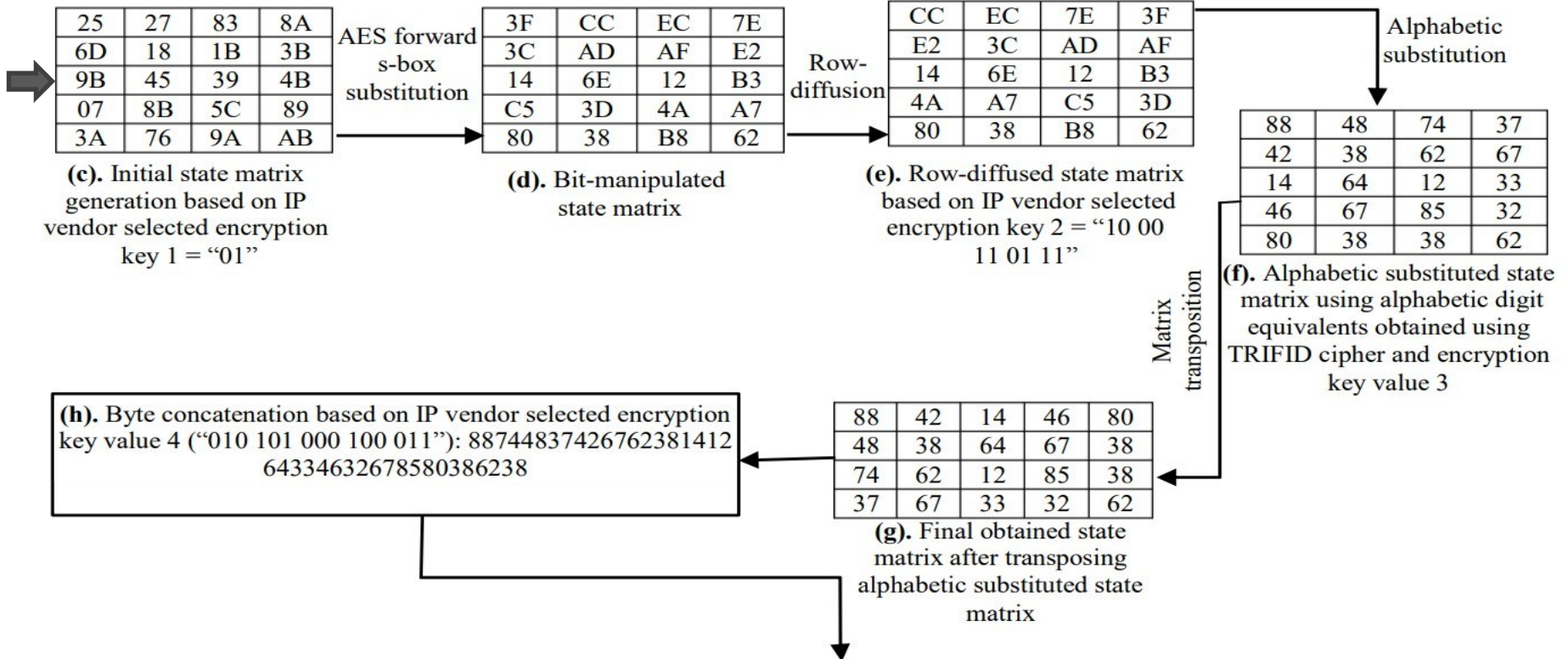
	C0	C1	C2	C3	C4	C5	C6
Red(R)	R0	R22	R22	R32	R34	R35	R36
Green (G)	R1	R23	R23	R33	R39	R40	R41
Indigo (I)	R2	R24	R24	R37	-	-	-
Blue (BL)	R3	R25	R25	R38	-	-	-
Yellow (Y)	R4	R26	R26	R26	R26	R26	-
Black (B)	R5	-	R27	-	-	-	-
Violet (V)	R6	-	R28	-	-	-	-
Pink (P)	R7	-	R29	-	-	-	-
Lime (LI)	R13	R13	R13	R13	R13	-	-
Olive (O)	R8	R8	R30	-	-	-	-
Aqua (A)	R9	R31	R31	R31	R31	R31	-
Teal (T)	R10	R10	-	-	-	-	-
Gray (G)	R11	R11	-	-	-	-	-
Maroon (M)	R12	R12	-	-	-	-	-
Silver (S)	R14	R14	-	-	-	-	-
Khaki (K)	R15	R15	-	-	-	-	-
Lavender (L)	R16	R16	-	-	-	-	-
Crimson (C)	R17	R17	-	-	-	-	-
Wheat (W)	R19	R19	-	-	-	-	-
Beige (B)	R18	R18	R18	R18	R18	-	-
Magenta (M)	R20	-	-	-	-	-	-
Orange (O)	R21	-	-	-	-	-	-

(a). Secret data extracted from register allocation table of sharpening filter scheduled using PSO-based architectural exploration resource constraints (4 adders and 2 multipliers):(0,36),(32,34),(33,39),(35,36),(1,40),(25,38),(7,29),(3,25),(22,32),(40,41),(22,35),(23,40),(8,30),(0,35),(0,32),(32,36),(1,33),(1,39),(33,41),(39,41),(4,26),(23,33),(22,34),(34,36),(23,39),(5,27),(0,22),(9,31),(1,23),(32,35),(0,34),(33,40),(2,37),(24,37),(6,28),(1,41),(39,40),(34,35),(22,36),(23,41),(3,38),(2,24)



(b). Conversion of secret data extracted into their corresponding hexadecimal format : (0,7), (0,9), (0,D), (0,E),(2,5),(2,7),(8,3),(8,A),(2,9),(0,6),(7,5),(2,4),(6,D),(1,8),(1,B),(3,B),(0,2),(8,0),(9,7),(1,9),(9,B),(4,5),(3,9),(4,B),(7,4),(5,6),(7,E),(0,5),(0,7),(8,B),(5,C),(8,9),(2,6),(0,4), (1,3),(1,A),(3,A),(7,6), (9,A), (A,B),(4,6)

Demonstration continued:



Trifid cipher computation and alphabetic substitution:

- Computing TRIFID cipher on "A":
- Let IP vendor selected key: EDRFTV\$QAWSZMXNCBGYHUJIKOLP
- Here, row number (a) is 3, column number (b) is 3, and square matrix (c) number is 1. The state corresponding to "A" is 331. Similarly, the state corresponding to the remaining alphabets is computed based on chosen key.

Square matrix 1			Square matrix 2			Square matrix 3		
E	D	R	W	S	Z	Y	H	U
F	T	V	M	X	N	J	I	K
\$	Q	A	C	B	G	O	L	B

Table 2: Square matrix representation of the key for TRIFID cipher computation corresponding to alphabet A

Table 3: Final obtained digit equivalents after alphabetic substitution

Assumed key	Alphabet	Corresponding TRIFID cipher state	Defined rule	Output
100	A	331	$a*b*c$	6
010	B	122	$a+b+c$	3
101	C	222	$ a-b-c $	8
011	D	233	$ a-b+c $	2
000	E	212	$(a+b)/c$	4
001	F	313	$(a+b)*c$	7

Generation of final signature and its corresponding hardware security constraints :



(i). Final generated digital signature: 10001000111100100100011111100101101111101011100011001
10110100111110011011101111100 01011000011100011010111000



(j). Final generated hardware security constraints based on IP vendor selected encoding rule:
(R0,R2),(R0,R4),---,(R2,R38),(R2,R40),(R4,R6) ,(R4,R8),---,(R6,R12),(R6,R14),(R1,R3),(R1,R5),---
,(R1,R39),(R1,R41),(R3,R5),(R3,R7),---,(R5,R23),(R5,R25)

Mapping rule: The generated signature is mapped to its corresponding hardware security constraints as per the IP vendor selected mapping rule (if encoding bit of signature is '0' then embed an edge between (even, even) storage variable pair, otherwise embed an edge between (odd, odd) storage variable pair).

Final register allocation table after embedding of security constraints :



Table 4: Register allocation table (both before and after) in one table:

	C0	C1	C2	C3	C4	C5	C6
Red(R)	R0	R22	R22	R32	R34	R35	R36
Green (G)	R1	R23	R23	R33	R39	R40	R41
Indigo (I)	R2	R24	R24	R37	-	-	-
Blue (BL)	R3	R25	R25/R28	R38	R34	-	R36
Yellow (Y)	R4	R26	R26/R27	R26	R26	R26	-
Black (B)	R5	R26	R27/R26	R26	R26	R26	-
Violet (V)	R6	R25	R28/R25	-	-	-	-
Pink (P)	R7	-	R29	-	-	-	-
Lime (LI)	R13	R13	R13	R13	R13	-	-
Olive (O)	R8	R8	R30	-	-	-	-
Aqua (A)	R9	R31	R31	R31	R31	R31	-
Teal (T)	R10	R10	-	-	-	-	-
Gray (G)	R11	R11	-	-	-	-	-
Maroon (M)	R12	R12	-	-	-	-	-
Silver (S)	R14	R14	-	-	-	-	-
Khaki (K)	R15	R15	-	-	-	-	-
Lavender (L)	R16	R16	-	-	-	-	-
Crimson (C)	R17	R17	-	-	-	-	-
Wheat (W)	R19	R19	-	-	-	-	-
Beige (B)	R18	R18	R18	R18	R18	-	-
Magenta (M)	R20	R22	R22	R32	-	-	-
Orange (O)	R21	R24	R24	-	-	-	-

Evaluation parameters:

➤ **Evaluation of Robustness Using Probability of Coincidence (Pc):**

$$P_c = \left(1 - \frac{1}{c}\right)^f$$

‘c’ denotes the number of registers used in the RAT and ‘f’ denotes the number of hardware constraints added.

➤ **Evaluation of tamper tolerance (TT):**

$$TT = (w)^f$$

‘w’ is the number of types of digits in the signature and ‘f’ is the signature size (or the number of corresponding hardware security constraints)

➤ **Design cost:**

$$Design\ cost = q_1 * \left(\frac{Area(A) - A_{con}}{A_{max}}\right) + q_2 * \left(\frac{Latency(L) - L_{con}}{L_{max}}\right)$$

where $q_1=0.5$ and $q_2=0.5$ are designer-defined weighing factors used to provide equal weightage to design area (A) and execution time (latency (L)) during design cost function evaluation. Further, A_{max} and L_{max} represents maximum design area (determined with available maximum functional resources) and time (delay) (determined with available minimum functional resources)

Results:



Table 5: Area, latency, cost, and resource configuration of proposed hardware security methodology before and after embedding signature

Benchmarks	Explored resource configuration	Baseline unsecured design (before signature embedding)			Signature embedded secured design			Design cost overhead %
		Design area (um ²)	Design latency (ps)	Design cost	Design area (um ²)	Design latency (ps)	Design cost	
Blur filter (BF)	3(+), 1(*)	147.849	927.399	-0.112	147.849	927.399	-0.112	0
Sharpening filter (SF)	4(+), 2(*)	243.79	794.91	-0.193	243.79	794.91	-0.193	0
Laplace edge detection filter (LED)	2(+), 2(*)	199.75	728.671	-0.108	199.75	728.671	-0.108	0
Vertical embossment (VE)	1(+), 1(*)	99.09	596.185	-0.077	99.09	596.185	-0.077	0
Horizontal embossment (HE)	1(+), 1(*)	99.09	596.185	-0.077	99.09	596.185	-0.077	0

Results:

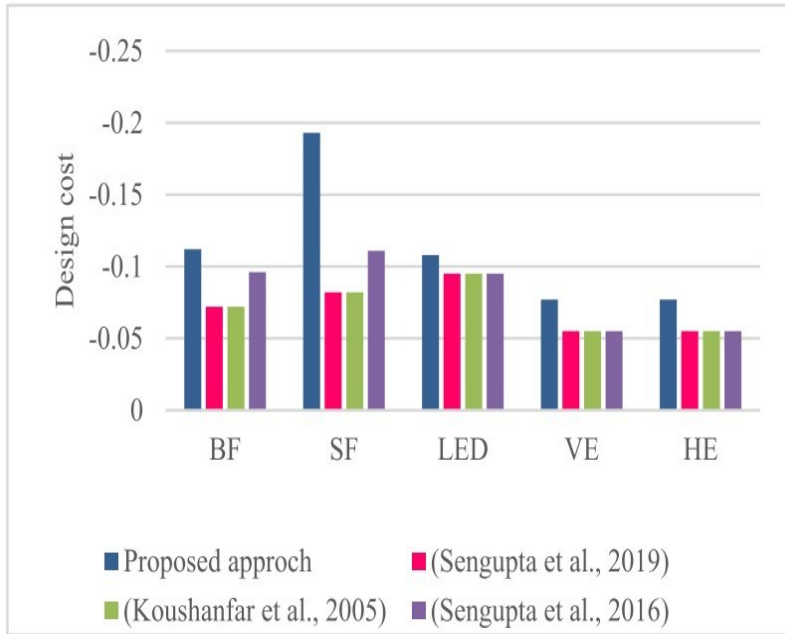


Figure 5: Design cost comparison between proposed approach and (Sengupta et al., 2019), (Koushanfar et al., 2005), and (Sengupta et al., 2016)

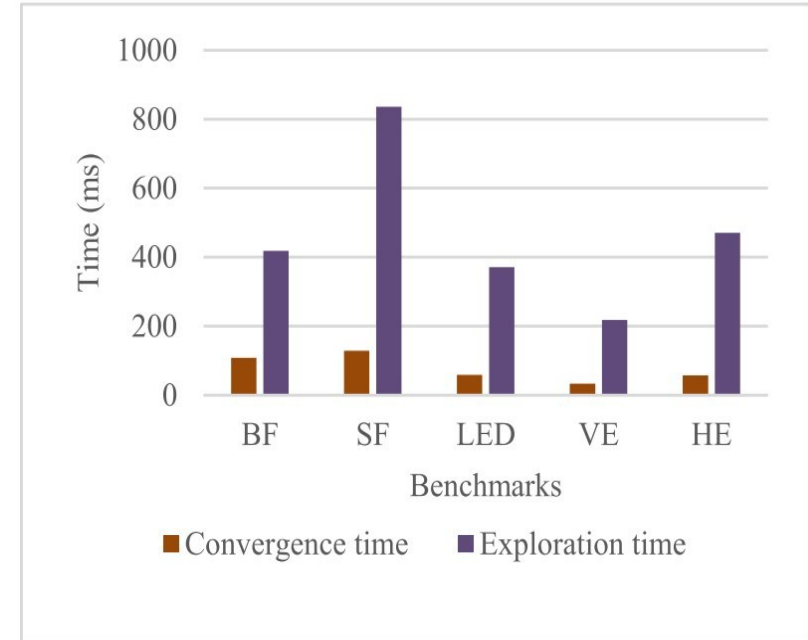


Figure 6: Convergence time and exploration time for the proposed methodology

Results (Pc comparison):

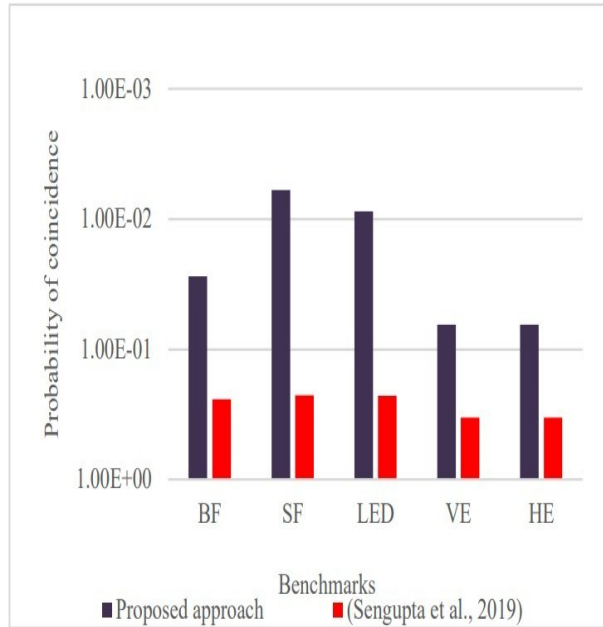


Figure 7: Comparison of probability of coincidence between the proposed and (Sengupta et al., 2019)

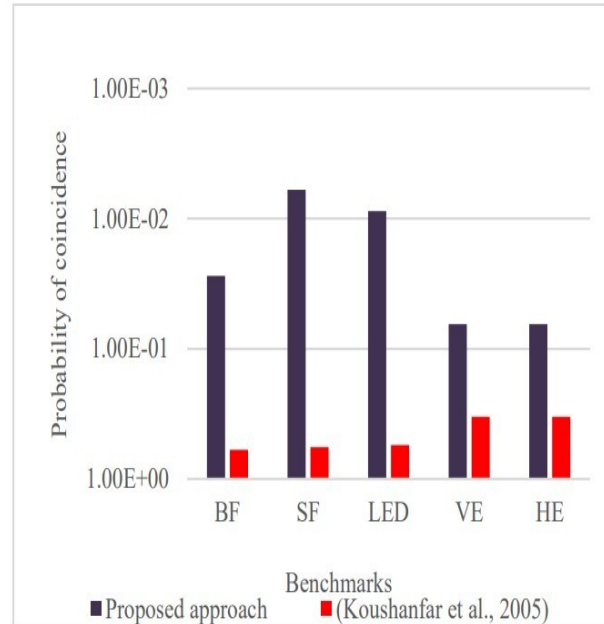


Figure 8: Comparison of probability of coincidence between the proposed and (Koushanfar et al., 2005)

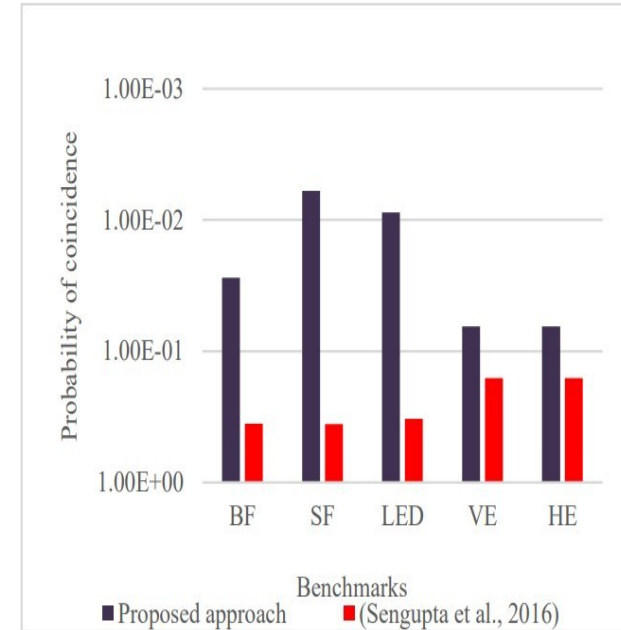


Figure 9: Comparison of probability of coincidence between the proposed and (Sengupta et al., 2016)

Results (TT comparison):

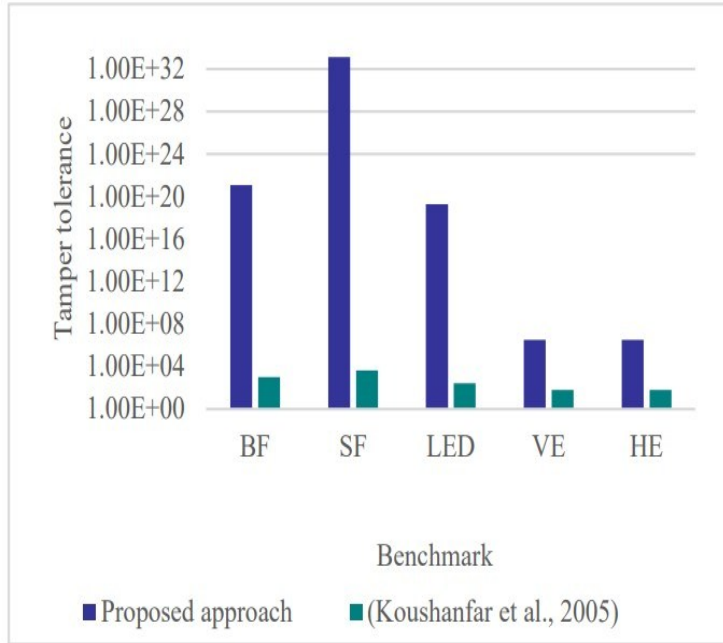


Figure 10: Comparison of tamper tolerance between the proposed and (Koushanfar et al., 2005)

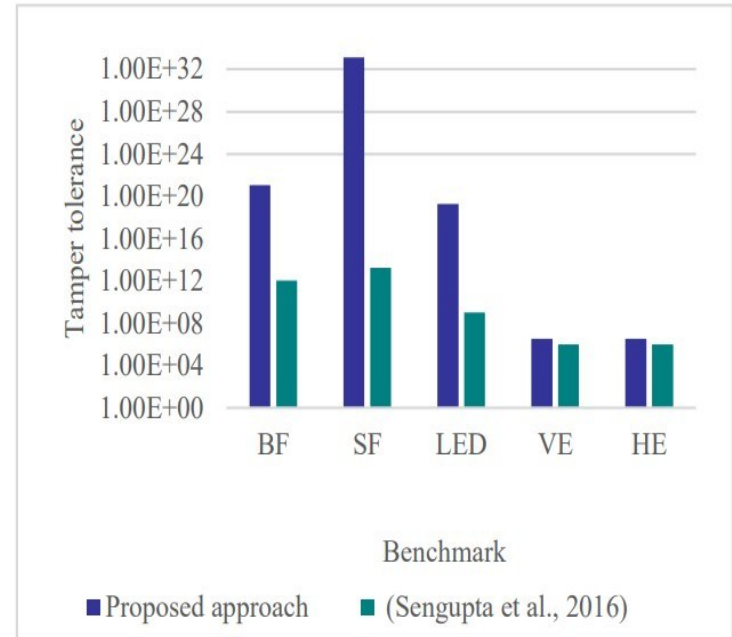


Figure 11: Comparison of tamper tolerance between the proposed and (Sengupta et al., 2016)

Results (optimality analysis):

Table 6: Result of the proposed approach in terms of optimality

Benchmarks	G	Spacing (S)	Spread (S_D)	Weighted metric (W_s)
BF	0.00	0.111	0.372	0.186
SF	0.00	0.745	0.611	0.306
VE	0.00	0.00	0.00	0.00
HE	0.00	0.00	0.00	0.00
LED	0.00	0.5	0.167	0.084

- A zero (or slightly higher value than zero) value for all metrics indicates that all solutions generated through the proposed approach lie on the true Pareto front and the uniform distribution of Pareto points on the curve.

Conclusion and future work:



- A novel **PSO driven multi-phase encryption based security methodology** for designing optimal secured image processing filter IP cores has been presented. The proposed PSO-driven multi-phase encryption mechanism employs strong security layers such as bit manipulation, row diffusion, TRIFID cipher computation, alphabetic substitution, and byte concatenation to generate a highly robust and tamper-tolerant signature.
- The proposed approach currently does not incorporate the capability to generate optimal datapath for loop based applications (such as FIR filter) using PSO based architecture exploration.
- The proposed approach uses register allocation table (RAT) of target image processing application to generate secret data, which is further used to generate hardware security constraints. This may lead to sometime generation of limited number of security constraints.

Publication list (Journals):



Journal publications:

1. A. Sengupta, R. Chaurasia and **A. Anshul**, "Robust Security of Hardware Accelerators Using Protein Molecular Biometric Signature and Facial Biometric Encryption Key," in **IEEE Transactions on Very Large Scale Integration (VLSI) Systems**, doi: 10.1109/TVLSI.2023.3265559, **Impact Factor: 2.77**.
2. M. Rathor, A. Sengupta, R. Chaurasia and **A. Anshul**, "Exploring Handwritten Signature Image Features for Hardware Security," in **IEEE Transactions on Dependable and Secure Computing**, 2022, doi: 10.1109/TDSC.2022.3218506, **Impact factor 6.79**.
3. **Aditya Anshul**, Anirban Sengupta, PSO based exploration of multi-phase encryption based secured image processing filter hardware IP core datapath during high level synthesis, **Elsevier Expert Systems with Applications**, Volume 223, 2023, 119927, ISSN 0957-4174, <https://doi.org/10.1016/j.eswa.2023.119927>, **Impact Factor: 8.66**.
4. Mahendra Rathor, **Aditya Anshul**, K Bharath, Rahul Chaurasia, Anirban Sengupta, Quadruple phase watermarking during high level synthesis for securing reusable hardware intellectual property cores, **Elsevier Computers and Electrical Engineering**, Volume 105, 2023, 108476, ISSN 0045-7906, <https://doi.org/10.1016/j.compeleceng.2022.108476>, **Impact factor: 4.15**.
5. R. Chaurasia, **A. Anshul**, A. Sengupta and S. Gupta, "Palmprint Biometric Versus Encrypted Hash Based Digital Signature for Securing DSP Cores Used in CE Systems," in **IEEE Consumer Electronics Magazine**, vol. 11, no. 5, pp. 73-80, 1 Sept. 2022, doi: 10.1109/MCE.2022.3153276, **Impact Factor: 4.13**.

Publication list (Conferences):



Conference publications:

1. **A. Anshul**, K. Bharath and A. Sengupta, "Designing Low Cost Secured DSP Core using Steganography and PSO for CE systems," 2022 **IEEE International Symposium on Smart Electronic Systems (iSES)**, Warangal, India, 2022, pp. 95-100, doi: 10.1109/iSES54909.2022.00030.
2. **A. Anshul** and A. Sengupta, "IP Core Protection of Image Processing Filters with Multi-Level Encryption and Covert Steganographic Security Constraints," 2022 **IEEE International Symposium on Smart Electronic Systems (iSES)**, Warangal, India, 2022, pp. 83-88, doi: 10.1109/iSES54909.2022.00028.

Publication list (Book Chapters):

Book Chapters:

1. Anirban Sengupta, **Aditya Anshul** "Palmprint Biometrics Vs. Fingerprint Biometrics Vs. Digital Signature using Encrypted Hash: Qualitative and Quantitative Comparison for Security of DSP coprocessors", **IET Book** "Physical Biometrics for Hardware Security of DSP and Machine Learning Coprocessors", 2023.
2. Anirban Sengupta, **Aditya Anshul** "Secured Design Flow using Palmprint Biometrics, Steganography and PSO for DSP coprocessors", **IET Book** "Physical Biometrics for Hardware Security of DSP and Machine Learning Coprocessors", 2023.
3. Anirban Sengupta, **Aditya Anshul** "Taxonomy of Hardware Security Methodologies: IP Core Protection and Obfuscation", **IET Book** "Physical Biometrics for Hardware Security of DSP and Machine Learning Coprocessors", 2023.
4. **Aditya Anshul**, Rahul Chaurasia, Anirban Sengupta "Securing Hardware Coprocessors against Piracy using Biometrics for Secured IoT systems", **IET Book** "Artificial Intelligence for Biometrics and Cybersecurity", 2023.
5. **Aditya Anshul**, Anirban Sengupta "Role of Consumer Technology and Connected Electronic Devices on SCM: A Discussion on its Usages, Impact, and Challenges", **UTHM Book** "Evolution of Information, Communication and Computing System", 2023.

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1. A. Sengupta, R. Chaurasia and A. Anshul, "Robust Security of Hardware Accelerators Using Protein Molecular Biometric Signature and Facial Biometric Encryption Key," in IEEE Transactions on Very Large Scale Integration (VLSI) Systems, doi: 10.1109/TVLSI.2023.3265559.
2. M. Rathor, A. Sengupta, R. Chaurasia and A. Anshul, "Exploring Handwritten Signature Image Features for Hardware Security," in IEEE Transactions on Dependable and Secure Computing, 2022, doi: 10.1109/TDSC.2022.3218506.
3. Aditya Anshul, Anirban Sengupta, PSO based exploration of multi-phase encryption based secured image processing filter hardware IP core datapath during high level synthesis, *Elsevier Expert Systems with Applications*, Volume 223, 2023, 119927.
4. C. Pilato, S. Garg, K. Wu, R. Karri and F. Regazzoni, "Securing Hardware Accelerators: A New Challenge for High-Level Synthesis," *IEEE Embedded Systems Letters*, vol. 10, no. 3, pp. 77-80, Sept. 2018.
5. D. Tsiktsiris, D. Ziouzos, and M. Dasygenis, "A portable image processing accelerator using FPGA," in Proc. *MOCAS*T, 2018, pp. 1–4.
6. C. Shu, W. Pang, H. Liu, and S. Lu, "High energy efficiency FPGA based accelerator for convolutional neural networks using weight combination," in Proc. *ICSIP*, Wuxi, China, 2019, pp. 578–582.

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7. F. Koushanfar, I. Hong, and M. Potkonjak, "Behavioral synthesis techniques for intellectual property protection," *ACM Trans. Design Autom. Electron. Syst.*, vol. 10, no. 3, pp. 523–545, Jul. 2005.
8. A. Sengupta and M. Rathor, "IP Core Steganography for Protecting DSP Kernels Used in CE Systems," in *IEEE Transactions on Consumer Electronics*, vol. 65, no. 4, pp. 506-515, Nov. 2019.
9. A. Sengupta and S. Bhadauria, "Exploring Low Cost Optimal Watermark for Reusable IP cores During High Level Synthesis," *IEEE Access*, vol. 4, pp. 2198–2215, 2016.
10. Vipul Kumar Mishra, Anirban Sengupta, MO-PSE: Adaptive multiobjective particle swarm optimization based design space exploration in architectural synthesis for application specific processor design, *Advances in Engineering Software*, Volume 67, 2014, Pages 111- 124.
11. V. Krishnan and S. Katkoori, "A genetic algorithm for the design space exploration of datapaths during high-level synthesis," *IEEE Transactions on Evolutionary Computation*, vol. 10, no. 3, pp. 213-229, June 2006.
12. A. Sengupta and S. Bhadauria, "Automated exploration of datapath in high level synthesis using temperature dependent bacterial foraging optimization algorithm," 2014 *IEEE 27th Canadian Conference on Electrical and Computer Engineering (CCECE)*, 2014, pp. 1-5.



Thank You!